📘 SESSION 6 – LOGISTIC REGRESSION WITH MULTIPLE INPUTS

🎯 Topic

- Logistic Regression with multiple features
- Real-world dataset understanding
- Feature impact on prediction
- Probability interpretation
- Practical implementation

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1. 🎯 Learning Objectives

By the end of this session, students will be able to:

- Understand Logistic Regression with **multiple inputs (features)**
- Identify how different features affect prediction
- Build a **realistic classification model**
- Interpret coefficients and probabilities
- Handle structured dataset like real industry data

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2. 🔁 Quick Recap (Connect Previous Session)

Start class like this:

"In the last session, we used Logistic Regression with only one input — study hours.
But in real life, decisions are never based on a single factor.
Today we will use multiple inputs to make better predictions."

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3. 📖 Why Multiple Inputs?

Real-Life Thinking

Example: Student Pass/Fail

Do we decide result only on:

- Study hours? ❌

Or also:

- Attendance ✔
- Previous marks ✔
- Practice ✔

👉 That's why we need multiple features.

Teaching Line

"More relevant features → better prediction."

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4. 📊 Logistic Regression with Multiple Inputs

Equation Idea (Simple)

Then: z=b0+b1x1+b2x2+b3x3

$$y = sigmoid(z)$$

Meaning

- Each feature has its own weight
- Model combines all features
- Then applies sigmoid

Teaching Line

"Each feature contributes to the final decision."

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5. 📌 Example Dataset (Real-Life)

Study Hours	Attendance	Previous Marks	Result
2	60	50	Fail
6	80	70	Pass

Model learns:

- High study + good attendance → Pass
- Low study + low attendance → Fail

6. 📊 Understanding Feature Importance

Concept

Each feature has a **coefficient**

- Positive coefficient → increases probability
- Negative coefficient → decreases probability

Example

Feature	Effect
Study Hours	Positive
Attendance	Positive
Low marks	Negative

"Coefficient tells how important a feature is in prediction."

Decision Boundary (Now Multi-Dimensional)

Earlier:

- Straight line (1 feature)

Now:

- Plane or hyperplane

👉 Students don't need deep maths — just intuition.

8. 📊 Probability Interpretation

Model gives:

$$P(\text{Pass}) = 0.85$$

👉 Meaning:

"85% chance student will pass"

"Logistic Regression always thinks in probabilities."

9. 📊 Real-Life Use Cases

- Loan approval (income + credit score + age)
- Medical diagnosis (multiple symptoms)
- Hiring decision (skills + experience + education)

10. 🎯 Interview Concepts

Q1: Why multiple inputs?

👉 Real-world problems depend on multiple factors.

Q2: What does coefficient mean?

👉 Impact of feature on prediction.

Q3: Can Logistic Regression handle multiple features?

👉 Yes, easily.

Q4: What happens if we add irrelevant features?

👉 Model may become less accurate.

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11. 💻 PRACTICAL (YOUR FORMAT)

🎯 Objective

Predict student result using:

- Study Hours
- Attendance
- Previous Marks

12. 📊 Output Interpretation

Explain clearly:

Observation:

- Model considers all 3 features
- Probability changes based on combination

Example:

Feature Combination	Result
Low study + low attendance	Fail
High study + high marks	Pass

Conclusion:

“Model becomes more accurate using multiple inputs”